Birth–Death Process

Queueing Theory, Nuclear Fission, and the NEXUS-1 Kinetics Engine

This note develops the theoretical bridge between continuous-time Markov chains — specifically the Birth–Death process from classical queueing theory — and the stochastic model of neutron population dynamics in a nuclear reactor. It then shows how the deterministic six-group point-kinetics solver implemented in NEXUS-1 emerges as the mean-field limit of that stochastic process, and discusses what each formulation can and cannot claim.

Document scope

Theoretical companion to the NEXUS-1 demonstrator (build NX1-094C79E0A366D34F). All physics constants are representative textbook-order values. This document does not constitute a safety case or a validated model.

Author: Agathangelidis Grigorios

Build: NX1-094C79E0A366D34F

1 Queueing Theory and the Birth–Death Process

Queueing theory studies systems in which entities — customers, jobs, packets — arrive, wait in a queue, receive service, and depart. The simplest family of such models is the M/M/1 queue and its generalisations, all of which rest on a single probabilistic backbone: the Birth–Death (BD) process.

1.1 The Continuous-Time Markov Chain

A BD process is a continuous-time Markov chain (CTMC) on the non-negative integers {0, 1, 2, ...}. The system state n represents the number of entities currently in the system. Transitions are restricted to nearest neighbours:

- $n \to n+1$ at rate λ_n (a birth — one entity arrives or is created)
- $n \to n-1$ at rate μ_n (a death — one entity departs or is removed)

The state-dependent rates λ_n and μ_n fully characterise the process. No jumps of size 2 or larger are allowed — the chain moves in unit steps. Below is the canonical state diagram:

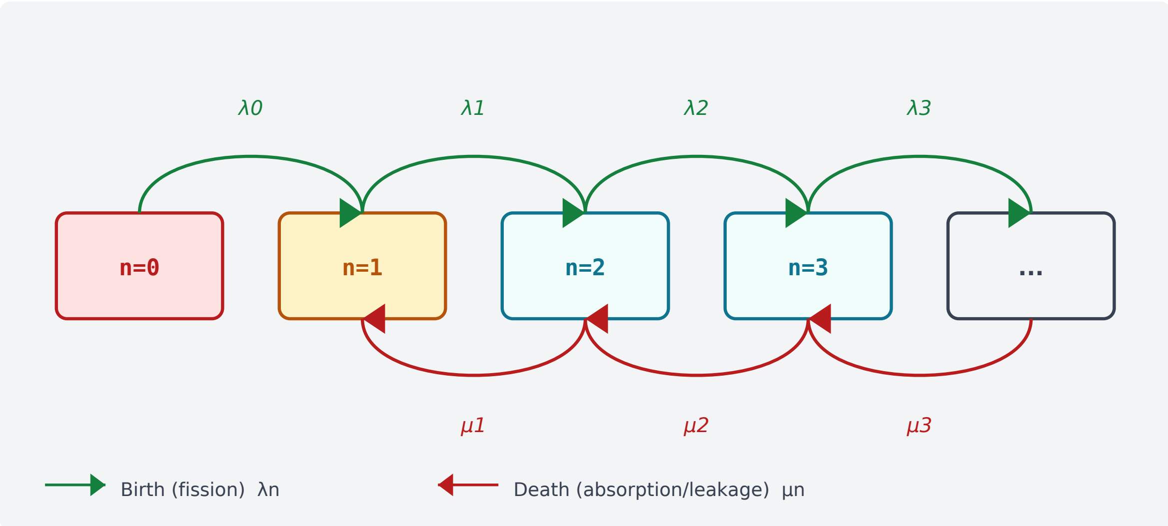

Figure 1 — Birth–Death chain. Green arcs: births (fissions). Red arcs: deaths (absorption / leakage). State n = 0 is absorbing.

1.2 Steady-State Distribution

When a steady-state distribution exists (requires λ_n and μ_n to satisfy certain stability conditions), it is given by the detailed balance equations:

$$\lambda_n \pi_n = \mu_{n+1} \pi_{n+1} \text{ for all } n = 0, 1, 2, \ldots$$

Unrolling from state 0:

$$\pi_n = \pi_0 \cdot (\lambda_0 \lambda_1 \ldots \lambda_{n-1}) / (\mu_1 \mu_2 \ldots \mu_n)$$

For the homogeneous case $\lambda_n = \lambda$ and $\mu_n = \mu$, the ratio $\rho = \lambda/\mu$ is the traffic intensity. The queue is stable if and only if $\rho < 1$.

1.3 Transient Behaviour and the Master Equation

For time-varying state probabilities $P_n(t) = \text{Prob}\{N(t) = n\}$, the dynamics are governed by the Kolmogorov forward equations, also called the master equation:

$$dP_n/dt = \lambda_{n-1} P_{n-1}(t) + \mu_{n+1} P_{n+1}(t) - (\lambda_n + \mu_n) P_n(t)$$

This is an infinite-dimensional system of coupled ODEs — one equation per state. The mean $\langle N(t)\rangle = \Sigma\, n\, P_n(t)$ obeys a simpler equation that, under certain conditions, decouples from the full distribution. That decoupling is the key step connecting the stochastic and deterministic worlds.

2 Nuclear Fission as a Birth–Death Process

The neutron population in a reactor core is a natural candidate for the BD framework. At the microscopic scale, individual neutrons are born (by fission) and die (by absorption or leakage) stochastically. The mapping is exact:

BD concept	Nuclear fission mapping
`State n`	Neutron population in the reactor at time t
`Birth n -> n+1`	A neutron induces fission; approximately 2.4 prompt neutrons emitted per event
`Birth rate lambda_n`	lambda_n = n * nu * Sigma_f * v (fission rate per neutron)
`Death n -> n-1`	Neutron absorbed in fuel, moderator, or control rod — or lost by leakage
`Death rate mu_n`	mu_n = n * (Sigma_a + Sigma_leak) * v (removal rate per neutron)
`State n = 0`	Chain extinct — absorbing state; no spontaneous neutron rebirth possible
`rho = lambda/mu`	Effective multiplication factor k_eff = nu * Sigma_f / (Sigma_a + Sigma_leak)
rho < 1 (stable)	k < 1 — sub-critical; neutron population decays exponentially to zero
rho = 1 (boundary)	k = 1 — critical; chain reaction is self-sustaining at constant power
rho > 1 (unstable)	k > 1 — super-critical; exponential population growth

Table 1 — Exact mapping from queueing BD concepts to nuclear fission.

2.1 The k_{eff} Parameter as Traffic Intensity

In queueing theory the stability condition is $\rho = \lambda/\mu < 1$. In reactor physics the exactly analogous quantity is the effective multiplication factor:

$$k_{eff} = \nu \, \Sigma_f \, / \, (\Sigma_a + \Sigma_{leak}) = \text{birth rate} \, / \, \text{death rate} = \rho$$

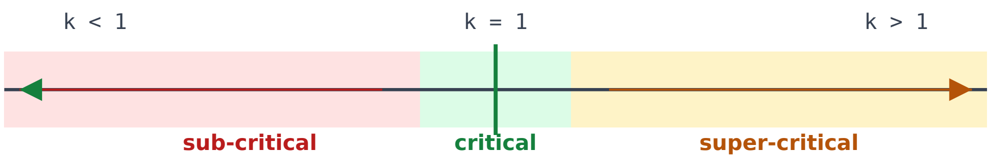

Figure 2 — k_{eff} as a traffic intensity. Sub-critical: queue drains to zero. Critical: balanced. Super-critical: queue grows without bound.

2.2 Delayed Neutrons as a Second Service Class

Approximately 0.65 % of fission neutrons are not emitted promptly. They are released over seconds to minutes by fission-product decay chains — the delayed neutrons. In the BD picture this introduces a two-class arrival process:

- Prompt neutrons: born within $\sim 10^{-14}$ s of fission, birth rate $\lambda_p = (1 - \beta)\, \nu\, \Sigma_f\, v$
- Delayed neutrons: born from 6 precursor groups, each with its own decay constant λ_i ($i = 1,\ldots,6$) and yield fraction β_i, with $\Sigma\beta_i = \beta \approx 0.0065$

The delayed fraction β is what makes reactors controllable: the effective reactor period is governed by the delayed timescale (~seconds), not the prompt timescale (~microseconds). Control rods operating on the second timescale are sufficient to govern the chain reaction as long as the reactor is held below prompt criticality ($k_p = k_{eff}(1-\beta) < 1$).

> n = 0 is absorbing. Unlike an empty queue that merely waits for the next customer to arrive, a reactor at n = 0 has no neutrons to initiate fission. The chain is extinct and cannot spontaneously restart. An external neutron source (AmBe or Cf-252 startup source) is physically inserted to re-initiate the chain — the analogue of an exogenous arrival injection.

3 Stochastic Analysis of the Neutron Population

The full stochastic description tracks the probability distribution over neutron populations rather than a single trajectory. Let $P_n(t)$ = Prob{N(t) = n}. The master equation for the pure-prompt BD model is:

$$dP_n/dt = \lambda_{n-1}P_{n-1} + \mu_{n+1}P_{n+1} - (\lambda_n+\mu_n)P_n$$

With linear (density-proportional) rates $\lambda_n = n\lambda$ and $\mu_n = n\mu$, this is a linear birth-death process. Its generating function $G(z,t) = \Sigma\, P_n(t)\, z^n$ satisfies:

$$\partial G/\partial t = (\lambda z - \mu)(z - 1)\ \partial G/\partial z$$

3.1 Exact Solution for the Linear Case

Starting from a single neutron (N(0) = 1), the exact solution gives extinction probability and conditional mean:

$$P_0(t) = \{\ \mu(e^{(\lambda-\mu)t}-1)\ \}\ /\ \{\ \lambda\ e^{(\lambda-\mu)t} - \mu\ \}\ \text{(extinction probability)}$$

$$E[N(t)\ |\ N(0)=1] = e^{(\lambda-\mu)t} = e^{(\omega_0)t}\ \text{(conditional mean)}$$

where $\omega_0 = \lambda - \mu = \mu(k_{eff} - 1)$ is the reactor period exponent. Three regimes:

Regime	Condition	Population behaviour	Extinction
Sub-critical	`k < 1, lambda < mu`	Mean decays exponentially to zero	Certain (P_0 -> 1)
Critical	`k = 1, lambda = mu`	Mean stays constant; random walk on states	Certain, but slow
Super-critical	`k > 1, lambda > mu`	Mean grows exponentially	Less than 1; survives with probability 1 - mu/lambda

Table 2 — Stochastic behaviour by criticality regime.

3.2 What the Stochastic Model Adds

The stochastic treatment provides quantities that a deterministic model cannot supply:

- Extinction probability — even a super-critical reactor has a non-zero probability of self-termination from fluctuations at low neutron numbers (relevant during startup).
- Variance and fluctuation spectra — the stochastic model predicts the variance Var[N(t)] = $e^{2\omega t}(e^{2\omega t}-1)(\lambda+\mu)/2\omega$, which grows far faster than the mean for k > 1.
- Reactor noise — subcritical noise measurements (Rossi-alpha, Feynman-Y experiments) are direct observables of λ and μ separately, not just their ratio k.
- Startup statistics — the probability distribution over which chain ignites a sustained reaction from a single source neutron follows directly from the BD extinction formula.

> The BD process also makes clear why $k_{eff} = 1$ is not a stable steady state in the probabilistic sense: at exact criticality the chain undergoes a random walk on the integers, and extinction is still certain in infinite time. A physical reactor avoids this by exploiting negative feedback (Doppler, moderator temperature coefficients) to act as a restoring force around the operating point — converting the Markov chain into a mean-reverting process.

4 The Deterministic Limit: Point-Kinetics

Taking the expectation of the master equation and applying a mean-field closure — replacing E[f(N)] with f(E[N]) — collapses the infinite-dimensional stochastic system into a finite-dimensional ODE. With six delayed-neutron groups this yields the point-kinetics equations:

$$dN/dt = [(k_{eff}(1-\beta) - 1)/\Lambda]\ N + \Sigma_{i=1}^{6}\ \lambda_i\ C_i$$

$$dC_i/dt = [\beta_i\ k_{eff}/\Lambda]\ N - \lambda_i\ C_i\ \ (i = 1,...,6)$$

Here N is the scalar neutron density (the mean of the stochastic N(t)), C_i is the concentration of the i-th delayed-neutron precursor group, Λ is the prompt neutron lifetime, and β_i / β are the delayed-neutron fractions.

4.1 Feedback Couplings

NEXUS-1 augments the bare point-kinetics equations with three reactivity-feedback terms that make k_{eff} a function of the reactor state:

$$k_{eff}(t) = k_0 + \rho_{ext}(t) + \alpha_D\ [T_f(t) - T_{f,0}] + \alpha_m\ [T_m(t) - T_{m,0}] + \rho_{Xe}(t)$$

The three physical feedback mechanisms:

- Doppler broadening — rising fuel temperature T_f broadens U-238 resonance capture cross-sections, increasing absorption and decreasing k. The Doppler coefficient α_D is negative (~−2.5 pcm/K), providing the fastest self-regulating mechanism.
- Moderator temperature coefficient — rising coolant/moderator temperature T_m decreases water density, reducing moderation efficiency and neutron thermalisation. α_m is negative in all Western PWR designs (~−40 pcm/K), the primary slow safety mechanism.
- Xenon-135 poisoning — Xe-135 (produced from Te-135 → I-135 → Xe-135 decay chain and directly from fission) has an enormous thermal absorption cross-section ($\sigma_a \approx 2.6 \times 10^6$ barns). Its concentration is tracked by a two-ODE iodine-xenon sub-model, producing the xenon transient after power changes and the post-trip xenon pit.

4.2 NEXUS-1 Solver Architecture

The NEXUS-1 point-kinetics solver (window.KIN.model) integrates the 8-dimensional ODE system [N, C_1,...,C_6, I, Xe] using an operator-splitting scheme:

- Stiff kinetics sub-step — prompt + delayed neutronics advanced with implicit Euler at a short timestep (Δt = 50 ms) to handle the fast prompt timescale.
- Thermal-hydraulics sub-step — fuel and moderator temperatures updated from a lumped thermal model driven by the current neutron power.
- Xenon / Iodine sub-step — the I-Xe ODE pair updated at the same timestep; xenon reactivity fed back into k_{eff} for the next step.

Two physics tiers in NEXUS-1. The Reactor Kinetics tab runs this genuine point-kinetics ODE solver. The Neutronics tab uses a separate, lighter illustrative model (display mappings driven by the fleet model) — clearly labelled on-screen. These two tiers must not be conflated.

5 Stochastic vs. Deterministic: Side-by-Side Comparison

The table below contrasts the two formulations across the dimensions most relevant to a reactor simulator:

Dimension	Stochastic BD process	Deterministic point-kinetics
State space	Probability distribution P(t) over integers {0, 1, 2, ...}	Scalar neutron density N(t), a single real number >= 0
Governing equation	Master equation (Kolmogorov forward) — one ODE per state, infinite-dimensional system	Point-kinetics ODEs — 8 coupled equations: N, 6 precursor groups C_i, I, Xe
Feedback	Rates lambda_n, mu_n become state-dependent; generally no closed-form solution	k_eff(t) is a function of fuel temp T_f, moderator temp T_m, and Xe via reactivity coefficients
Relationship	Fundamental and exact at all population levels	Mean-field limit: E[N(t)] satisfies the kinetics ODE when fluctuations are negligible relative to N
Extinction	P_0(t) computed exactly; extinction is certain even for k > 1 as t -> inf when starting from finite n	N(t) >= 0 always; no concept of extinction probability — deterministic trajectory only
Fluctuations	Var[N], reactor noise spectra, Rossi-alpha all follow naturally from the distribution	None — single deterministic trajectory; no variance information
Startup statistics	Ignition probability from a single source neutron: 1 - mu/lambda	Not accessible — must assume N(0) > 0; cannot describe ignition from a single neutron
Computational cost	High — Monte Carlo sampling or truncated master equation across thousands of states	Low — 8 ODEs, runs in-browser at 50 ms timestep in NEXUS-1
Validity regime	All populations: n = 0 through ~10^4 (source and intermediate range)	N >> 1 (full-power operation); approximation breaks down near extinction
NEXUS-1 use	Theoretical foundation — provides the stochastic justification for the kinetics model	Implemented as window.KIN.model; 6-group delayed-neutron solver with Doppler/Xe feedback
Observable link	Feynman-Y and Rossi-alpha noise experiments measure lambda and mu separately	Period measurements give k_eff / Lambda; reactivity perturbations give beta

Table 3 — Stochastic BD process vs. deterministic point-kinetics.

5.1 When the Mean-Field Closure Breaks Down

The deterministic model is an excellent approximation when the neutron population is large ($N \gg 1$) and fluctuations are a small fraction of the mean. This holds throughout normal power operation. The approximation fails in two regimes:

- Source range — during reactor startup from shutdown, N may be of order 10^3–10^6. Statistical fluctuations are significant; stochastic models or Monte Carlo neutron transport

(MCNP, OpenMC) are required for rigorous startup analysis.

- Near extinction — as the reactor approaches $n = 0$ after a trip, the deterministic model predicts smooth exponential decay while the stochastic model assigns a non-zero probability of the chain self-sustaining. This distinction matters for decay-heat and coolant calculations immediately post-trip.

NEXUS-1 operates in the high-power regime where point-kinetics is the appropriate tool. The stochastic BD formulation is retained as the intellectual foundation that justifies the structure of the kinetics equations — not as a competing model.

6 NEXUS-1 Representative Constants

The following table lists the six-group delayed-neutron parameters implemented in window.KIN.model. Values are representative of a Westinghouse PWR and are not tuned to a specific reactor. They are the numerical realisation of the BD birth rates (prompt) and the six-class delayed birth supplement.

Group i	Precursor half-life	Decay constant λ_i (s^{-1})	Yield fraction β_i	Physical origin
1	55.7 s	0.0124	0.000215	Br-87, Kr-89 precursors
2	22.7 s	0.0305	0.001424	Rb-93, Kr-91 family
3	6.22 s	0.111	0.001274	Y-97, Rb-92 region
4	2.30 s	0.301	0.002568	I-137, Cs-138 group
5	0.610 s	1.14	0.000748	Kr-88, Br-88 family
6	0.230 s	3.01	0.000273	He-5, Li-9 very short
Sum	–	–	**0.006502**	**Total delayed fraction beta**

Table 4 — Six-group delayed-neutron parameters in window.KIN.model (representative PWR values, Keepin 1965 basis).

Parameter	Symbol	Value	Notes
Prompt neutron lifetime	Lambda	20 us	PWR typical; governs prompt timescale
Doppler coefficient	alpha_D	-2.5 pcm/K	Negative in all PWRs; provides fast self-regulation
Moderator temp. coefficient	alpha_m	-40 pcm/K	Negative at power; primary slow feedback mechanism
Prompt fission neutrons	nu	2.43	Average neutron yield per U-235 fission event
Xenon-135 microscopic xs	sigma_Xe	2.65e6 b	Largest known thermal neutron absorption cross-section
Iodine-135 decay constant	lambda_I	2.87e-5 /s	Half-life 6.57 h
Xenon-135 decay constant	lambda_Xe	2.10e-5 /s	Half-life 9.17 h

Table 5 — Additional kinetics and feedback parameters implemented in NEXUS-1.

7 Conclusions

The Birth–Death process from queueing theory provides a rigorous stochastic foundation for the physics of nuclear fission chain reactions. The mapping is exact and instructive:

- Births = fission events; deaths = neutron removal. The traffic intensity ρ is the reactor's effective multiplication factor k_{eff}.
- The master equation governs the full probability distribution over neutron populations and is the correct description at all population levels.
- The deterministic six-group point-kinetics equations emerge as the mean-field limit: the ODE for the mean population E[N(t)] when fluctuations are negligible relative to N.
- Stochastic effects matter during reactor startup and near extinction; the deterministic model is appropriate — and computationally tractable — at power.
- NEXUS-1 implements the deterministic limit (window.KIN.model) with Doppler, moderator, and xenon feedback. The BD formulation is its theoretical parent.

The BD / queueing-theory connection is conceptually rich but deliberately kept outside the main NEXUS-1 companion book (From Grid to Core), whose intended reader does not require Markov-chain background. This document exists as a standalone theoretical note for readers with a stochastic-processes background who wish to understand the mathematical foundation of the kinetics engine.

Build reference

`NEXUS-1 demonstrator build: NX1-094C79E0A366D34F`

`Repository: github.com/gregory82gr/Nexus-1-phase-0`

Author: Agathangelidis Grigorios